

PAVAN GURURAJ SHIRAGUPPI

GEN AI ENGINEER | MLOPS & DEVOPS ENGINEER | WATSONX SME
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EDUCATION

Institute of Management Technology, Ghaziabad
MBA - Design Thinking and Entrepreneurship
07/2023 - 06/2024

Manipal Institute of Technology, Manipal
B.Tech - Electronics and Instrumentation
07/2015 - 06/2019

TECHNICAL SKILLS

GENERATIVE AI

LLM Fine-tuning RAG Architecture
Prompt Optimization LangChain Watsonx

RAGAS CrewAI

DEVOPS & INFRASTRUCTURE

Red Hat OpenShift Kubernetes Docker
Tekton Pipelines ArgoCD Ansible

Terraform Helm

MACHINE LEARNING

PyTorch TensorFlow Scikit-learn
XGBoost Hugging Face Transformers

DATA SCIENCE

Pandas NumPy Statistical Modeling
A/B Testing EDA Streamlit Gradio

NLP & COMPUTER VISION

BERT GPT Transformers OpenCV
Sentiment Analysis Text Classification

VECTOR DATABASES

ChromaDB Pinecone FAISS Weaviate

LANGUAGES

Python SQL Bash

AI INFRASTRUCTURE & DEPLOYMENT

AWS SageMaker MLflow Model Serving
LlamaIndex

CERTIFICATIONS

- Academy Accreditation - Databricks Lakehouse Fundamentals
- AMR Robot Training - Intermediate Level
- Japanese Language Proficiency - Level N2

PUBLICATIONS

RoboCafe: Multi-Robot Cafe Automation System
<https://youtu.be/cv0c5uoqdBo>

Library Robot: Autonomous Book Retrieval System
<https://youtu.be/0SNUPvxokJs>

Ritsumeikan University Robotics Demo
https://youtu.be/_y39o1qxa2Q

LANGUAGES

English
Native or Bilingual Proficiency

Japanese
Full Professional Proficiency

PROFILE

Generative AI Engineer and MLOps practitioner with a unique trajectory spanning Systems Integration, Data Science, and LLM Engineering. Proven track record of architecting and deploying production-grade AI infrastructure on IBM watsonx (.ai, .data, Orchestrate) and OpenShift, contributing to \$10M+ in revenue. Specialized in CI/CD pipelines for model deployment, LLM fine-tuning, RAG architectures, and enterprise DevOps practices. Expert in bridging AI development and platform operations at scale. Bilingual in English and Japanese (N2).

PROFESSIONAL EXPERIENCE

Watsonx Platform SME

01/2026 - Present

IBM INDIA PVT LTD | BENGALURU, INDIA

- Lead secure on-premise deployments of the full IBM watsonx stack, including watsonx.ai for LLM inference and Prompt Lab access, watsonx.data as a governed data lakehouse with Presto/Spark query engines, and watsonx Orchestrate for AI-powered workflow automation and digital worker orchestration across enterprise environments.
- Manage end-to-end deployment of the watsonx platform on Red Hat OpenShift, overseeing operator lifecycle management, pod scheduling, resource quotas, and persistent volume configuration to ensure high availability and compliance.
- Architect scalable, air-gapped watsonx.ai inference clusters with GPU node pools on OpenShift, enabling enterprise clients to run Granite and third-party LLMs on-premise with data residency guarantees.

Generative AI Engineer

08/2023 - 01/2026

IBM INDIA PVT LTD | BENGALURU, INDIA

- Spearheaded high-impact Generative AI engagements contributing to over \$10M in revenue, designing and optimizing enterprise LLM solutions for digital transformation.
- Fine-tuned and optimized custom Large Language Models (LLMs) including Llama, Granite, and GPT variants to maximize domain-specific accuracy and reduce token latency by 40%.
- Architected advanced RAG (Retrieval-Augmented Generation) systems integrating vector databases (ChromaDB, Pinecone) for context-aware AI applications.
- Built intelligent AI agents using IBM Watsonx Orchestrate and LangChain, enabling autonomous task execution and complex multi-step reasoning.

Generative AI Advisor, SME

02/2023 - 08/2023

KUSHAGRAMATI ANALYTICS | BENGALURU, INDIA

- Advised enterprise clients on Generative AI adoption strategies, identifying high-ROI use cases for LLM integration.
- Conducted feasibility studies for RAG architectures, guiding technical teams on model selection (OpenAI vs Open Source) and infrastructure requirements.
- Established governance frameworks for LLM evaluation, defining metrics for hallucination rates and response accuracy.

DevOps Engineer

04/2022 - 02/2023

HCI | OSAKA, JAPAN

- Designed and maintained CI/CD pipelines for robotic software systems, automating build, test, and deployment workflows to reduce release cycle time by 30%.
- Containerized AI inference services and robotic control modules using Docker, enabling consistent deployment across development and production environments.
- Managed Linux-based infrastructure for GPU compute nodes running NLP and computer vision workloads, ensuring high uptime and resource efficiency.
- Implemented monitoring and alerting for deployed ML models and robotic services using Prometheus and Grafana, enabling proactive incident response.

Systems Engineer

12/2020 - 03/2022

HCI | OSAKA, JAPAN

- Architected the RoboCafe bio-system, orchestrating interoperability between industrial arms (Kawasaki, Yaskawa, Mitsubishi) and service robots (Pudu Robotics, AGVs).
- Designed the precise integration protocols between stationary industrial robots and mobile service robots, enabling seamless physical handoffs and synchronized path planning.
- Integrated Computer Vision models with robot arm kinematics to enable autonomous object manipulation in a dynamic cafe environment.
- Bridged the gap between hardware interfaces (PLCs, Robot Controllers) and software AI modules, creating a seamless end-to-end automated solution.